



Scanctum

# Modular Web Application Security Scanner

A full-stack vulnerability detection system built for B.Tech IT (PBL) evaluation



# Problem Statement

## Web Application Vulnerabilities

Modern web apps face constant threats from XSS, SQL injection, and IDOR attacks

## Complex Security Tools

Existing scanners are expensive, difficult to configure, or lack real-time feedback

## Accessibility Gap

Students and developers need simple, affordable security testing solutions

# Project Objective

## 1 Build Modular Scanner

Create extensible security testing framework

## 2 Automated Detection

Identify vulnerabilities without manual intervention

## 3 Real-Time Updates

Provide live scan progress tracking

## 4 VAPT Reports

Generate professional vulnerability assessment documentation



# System Architecture



## Next.js Frontend

User interface and dashboard



## FastAPI Backend

REST API and task management



## PostgreSQL

Data persistence and results storage



## Redis + Celery

Task queue and WebSocket communication



## Scanner Engine

Vulnerability detection modules



# Technology Stack

## Backend & API

- FastAPI (Python)
- PostgreSQL (Database)
- Redis (Queue + Pub/Sub)
- Celery (Task processing)

## Frontend

- Next.js (React framework)
- TypeScript
- Tailwind CSS

## Infrastructure

- Docker (Containerization)
- WebSocket (Real-time updates)
- Async I/O (Scalability)



# Key Features



## Scan Modes

Quick Scan for fast results, Full Scan for comprehensive vulnerability assessment



## CVSS Risk Scoring

Industry-standard severity ratings using Common Vulnerability Scoring System



## Automated Reports

Generate PDF and JSON reports with evidence, severity, and remediation steps



## Scan Comparison

Compare multiple scan results to track security improvements over time



## Real-Time Tracking

Live progress updates via WebSocket connections



## Modular Design

Extensible architecture allows easy addition of new vulnerability detection modules

# How Scanning Works



## User Initiation

Specify target URL and scan configuration



## Task Creation

Backend generates scan tasks and stores in Redis queue



## Execution

Celery workers process tasks asynchronously



## Storage

Results saved to PostgreSQL with full details



## Visualization

Frontend displays findings with severity levels

# Vulnerability Detection

## Passive Analysis

- HTTP header inspection
- Cookie security checks
- Security misconfiguration detection

## Active Testing

- Payload injection testing
- Form parameter analysis
- Authentication bypass attempts



### XSS Detection

Cross-site scripting vulnerabilities

### SQL Injection

Database query manipulation

### IDOR

Unauthorized data access

### SSRF

Server-side request forgery



# Real-Time Updates & Reporting



## WebSocket Communication

Live progress streaming from server to frontend interface



## PDF/JSON Reports

Professional VAPT format with evidence and remediation guidance



## CVSS Integration

Automated severity scoring using industry-standard metrics



# Conclusion & Future Scope

## Project Status

- Backend and frontend completed
- Scanner engine fully functional
- Real-time system operational
- Docker deployment ready

## Future Enhancements

- AI-based vulnerability prediction
- CI/CD pipeline integration
- Cloud deployment options
- Auto-remediation suggestions

# Modular

System Design

Extensible architecture

# Full-Stack

Web Tool

Complete solution

# Scalable

Performance

Async processing